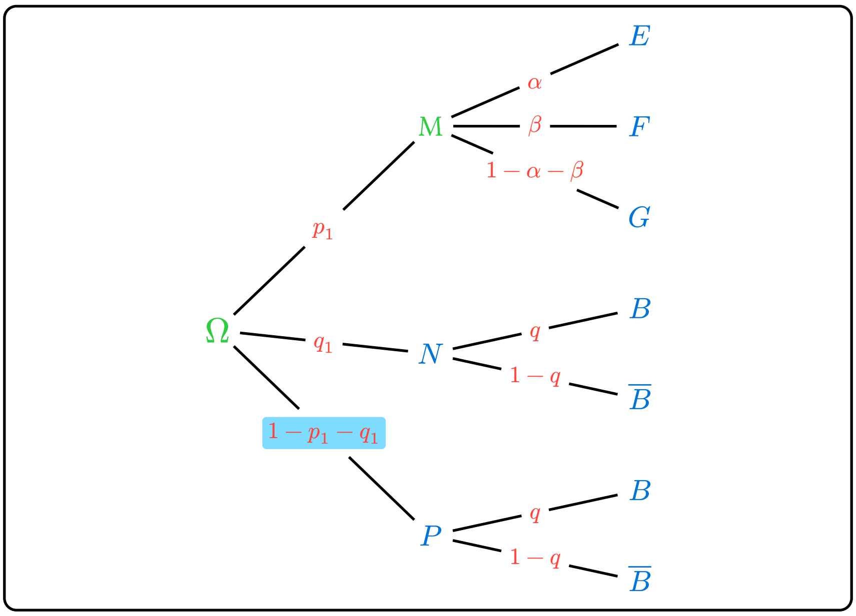
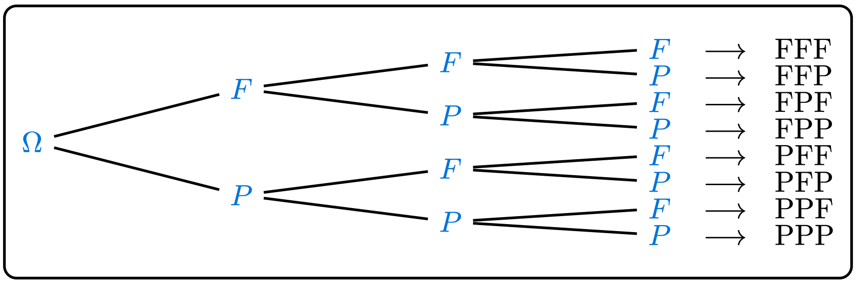
Probability Tree

– Quick reference –



– Overlaying custom labels (`extra`) –



- proba-tree()
- sn()
- sp()

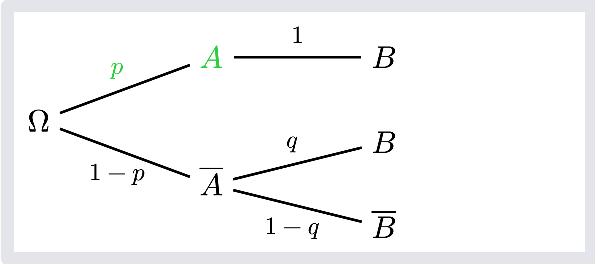
proba-tree

Draws a probability tree, growing from left to right.

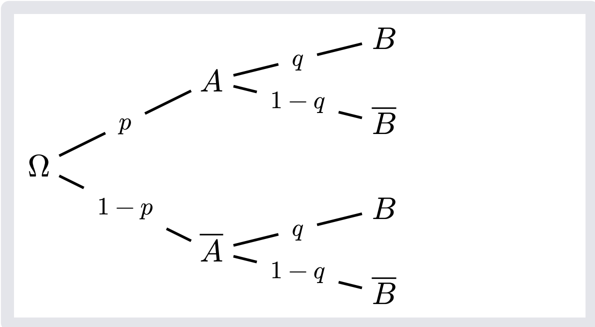
Each node of `data` is an array `(label, proba, ..children)`; the label and the proba can be raw content (`$A$`, `$p$`) or a local setting via [`sn`] / [`sp`].

```
#proba-tree(data: (
  [$Omega$],
  (sn($A$, style: (fill: green)), sp($p$,
style: (fill: green)), ([$B$], sp($1$,
position: "above"))),
  ([$overline(A)$], $1-p$, ([$B$], $q$),
```

```
([$overline(B)$], $1-q$)),
))
```




```
#proba-tree(proba-position: "on", proba-
padding: 5pt)
```



Parameters

```
proba-tree(
  h: float,
  v: float,
  proba-position: string,
  proba-distance: float,
  proba-sloped: bool,
  proba-padding: length,
  proba-style: dictionary,
  node-style: dictionary,
  first-child-top: bool,
  node-padding: float,
  data: array,
  extra: function
) -> content
```

h `float`

Horizontal extension of the edges (default: 2).

Default: `2.0`

v `float`

Vertical spread between branches (default: 0.8).

Default: `0.8`

proba-position `string`

Placement of the probabilities: `above`, `below`, `on` or `hybrid` (default: `hybrid`).

Default: `hybrid`

proba-distance `float`

Distance of the label from the edge (default: 0.35).

Default: `0.3`

proba-sloped `bool`

Aligns the probabilities along the edge, « sloped » (default: `false`).

Default: `false`

proba-padding `length`

In `on` mode, gap between the cut edge and the proba bbox (default: 3pt).

Default: `3pt`

proba-style `dictionary`

Global style of the probabilities. The `80%` size is ALWAYS kept as the base, unless redefined here.

Keys: `size`, `fill`, `weight`, `style`, `smallcaps`, `highlight`, `function` + `#text` params.

Default: `(size: 80%)`

node-style `dictionary`

Global style of the node texts, e.g. `(size: 11pt, fill: blue)` (default: `none`).

Keys: `size`, `fill`, `weight`, `style`, `smallcaps`, `highlight`, `function` + `#text` params.

Default: `none`

first-child-top `bool`

1st listed child at the top, left-to-right tree (default: `true`).

Default: `true`

node-padding `float`

Radius of the « hole » left around each letter (default: 0.3).

Default: `0.3`

data `array`

The tree to draw (last argument). Default: Ω root, A/Ā (p/1-p), B/B̄ (q/1-q).

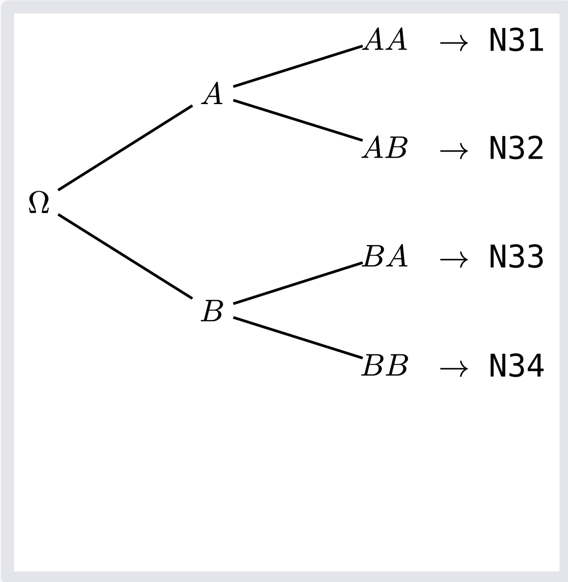
Default: `default-tree`

extra `function`

Callback drawn inside the canvas after the tree. It receives the node positions as a dictionary keyed `N<level><index>` (1-indexed), e.g. `N11` = root, `N41` = 1st leaf of a 4-level tree, plus the cetz `draw` namespace. Each position is a `(x, y)` pair in canvas units.

Indices are numbered in visual top-to-bottom order, whatever the value of `first-child-top`.

```
#proba-tree(
  data: (
    [$Omega$],
    ([$A$], $$, ([$A A$], $$), ([$A B$],
$$)),
    ([$B$], $$, ([$B A$], $$), ([$B B$],
$$)),
  ),
  extra: (pos, draw) => {
    for k in ("N31", "N32", "N33",
"N34") {
      let p = pos.at(k)
      draw.content((p.at(0) + 0.3,
p.at(1)), anchor: "west", [#h(1em)
$arrow.r$ #k])
    }
  },
)
```



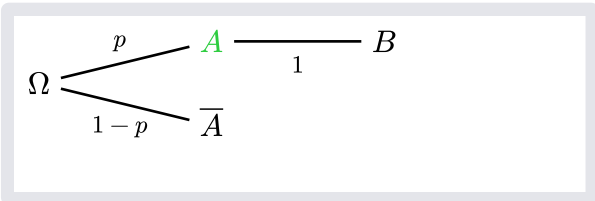
Default: none

sn

Locally sets a node: text and style.

Used as the label of a node in `data`, e.g. `(sn($A$, style: (fill: green)), $p$, ...)`. Any omitted parameter falls back to the global settings of [`proba-tree`].

```
#proba-tree(data: (
  [$Omega$],
  (sn($A$, style: (fill: green, weight:
"bold")), $p$, ([$B$], $1$)),
  ([$overline(A)$], $1-p$),
))
```



Parameters

```
sn(
  content: content,
  style: dictionary
) -> dictionary
```

content `content`

The node text (e.g. `$A$`, `[A]`, `"A"`).

style `dictionary`

Local style applied via `#text`, merged with `node-style`.

Keys: `size`, `fill`, `weight`, `style`, `smallcaps`, `highlight`, `function` + all `#text` parameters.

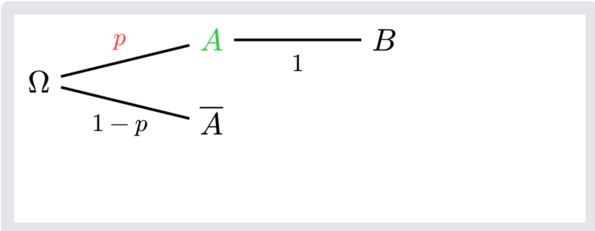
Default: `auto`

sp

Locally sets a probability: value and display options.

Used as the proba of a node in `data`, e.g. `([$A$], sp($p$, style: (weight: "bold")), ([$B$], $q$))`. Any omitted parameter falls back to the global settings of [`proba-tree`].

```
#proba-tree(data: (
  [$Omega$],
  (sn($A$, style: (fill: green)), sp($p$,
style: (fill: red, weight: "bold")),
([$B$], $1$)),
  ([$overline(A)$], $1-p$, ),
))
```



Parameters

```
sp(
  content: content,
  position: string,
  sloped: bool,
  distance: number,
  style: dictionary
) -> dictionary
```

content `content`

The displayed probability (e.g. `$p$`, `1-p`).

position `string`

Placement: `above`, `below`, `on` or `hybrid`.

Default: `auto`

sloped `bool`

Aligns the probability along the edge (« sloped »).

Default: `auto`

distance `number`

Distance of the label from the edge.

Default: `auto`

style `dictionary`

Local style applied via `#text`, merged with `proba-style`.

Keys: `size`, `fill`, `weight`, `style`, `smallcaps`, `highlight`,

`function` + all `#text` parameters.

Default: `auto`